JOB MARKET ANALYSIS

Project: PRDA – 04

Technologies: MySQL, SQL, Microsoft Power BI

Dataset: 742 Records and 42 Features

1. INTRODUCTION

The job market is continuously evolving due to technological advancements and increasing demand for data-driven decision-making. Data Science, Data Analytics, Machine Learning, and Artificial Intelligence have created many employment opportunities across different industries.

The objective of this project is to analyze job market data to understand employment trends, salary distributions, company hiring patterns, industry demand, and the skills required for various job roles.

This project uses SQL and Microsoft Power BI to analyze a dataset containing 742 job records and 42 features. SQL is used to retrieve, filter, aggregate, and analyze the data, while Power BI is used to create interactive visualizations and dashboards.

The analysis provides useful insights for job seekers, employers, and organizations to make informed career and recruitment decisions.

2.PROJECT OBJECTIVES

The main objectives of this project are:

1. To identify the states with the highest number of job opportunities.
2. To analyze minimum, maximum, and average salaries across different states.
3. To identify the top five industries offering Data Science-related jobs.
4. To find companies with the highest number of job openings.
5. To identify the most frequently available job titles.
6. To analyze salary distributions across different job titles.
7. To identify the most demanded technical skills by companies.
8. To examine the relationship between salary and educational qualifications.
9. To identify hiring trends and patterns across companies and industries.
10. To analyze different dataset features and derive meaningful insights.
11. To develop an interactive Power BI dashboard for effective data visualization.

3. DATASET DESCRIPTION

The project uses a Job Market Analysis dataset containing 742 records and 42 features collected from different job-related sources.

The dataset provides information about job titles, salary estimates, company details, locations, industries, sectors, company ratings, required skills, and educational qualifications.

Important dataset attributes:

Attribute	Description

Job Title	Name of the advertised job position
Salary Estimate	Estimated salary range
Job Description	Details of job responsibilities and requirements
Rating	Company rating
Company Name	Name of the hiring organization
Location	Location of the job
Headquarters	Company headquarters location
Size	Number of employees
Industry	Industry in which the company operates
Sector	Business sector
Revenue	Annual company revenue
Lower Salary	Minimum estimated salary

Upper Salary	Maximum estimated salary
Avg Salary (K)	Average salary in thousands of US dollars
Python, Spark, AWS, Excel	Technical skill indicators
Jobtitle_sim	Standardized job title
Seniority_by_title	Job seniority
Degree	Educational qualification indicator

The dataset also contains information about company ownership, founding year, competitors, hourly salary, and employer-provided salary details.

4. TOOLS AND TECHNOLOGIES

The following tools and technologies were used to complete this project:

1. MySQL: Used to store and manage the job market dataset.

2. SQL: Used to perform data extraction, filtering, aggregation, joins, grouping, and analytical queries.

3. Microsoft Power BI: Used to create charts, graphs, interactive dashboards, and visual representations of job market trends.

4. Microsoft Excel / CSV: Used as a data source and for preliminary data inspection, where applicable.

These tools helped transform raw job market data into meaningful information and business insights.

5. DATA CLEANING AND PREPROCESSING

Data cleaning and preprocessing were performed to improve the quality and reliability of the dataset before conducting analysis.

The following steps were considered:

1. Checked the dataset for missing and unavailable values.
2. Identified columns containing -1 values, which represent unavailable information.
3. Examined duplicate records and checked data consistency.
4. Verified the data types of important columns.
5. Checked salary-related columns for consistency.
6. Standardized job titles and company names wherever required.
7. Prepared the dataset for SQL-based analysis and Power BI visualization.

Missing Value Treatment:

The value -1 indicates that information was unavailable in the original dataset. These values were handled according to the data type and analytical requirements. For example, unavailable numerical values should not be treated as actual salaries or ratings. Categorical missing values can be classified as Unknown.

The cleaned data was then used for further analysis.

6.SQL ANALYSIS AND QUERIES

6.1 States with Most Number of Jobs

Objective: To identify the states offering the highest number of job opportunities.

SQL Approach: Grouped job records by state and counted the number of job postings. The results were sorted in descending order.

Finding: This analysis identifies the states with the highest concentration of job opportunities.

6.2 Average Minimum and Maximum Salaries in Different States

Objective: To compare minimum and maximum salary estimates across different states.

SQL Approach: Calculated the average lower salary and average upper salary for each state.

Finding: This analysis helps compare salary ranges across geographical locations.

6.3 Average Salary in Different States

Objective: To identify differences in average salaries across states.

SQL Approach: Grouped job postings by state and calculated the average salary.

Finding: This analysis helps identify states offering comparatively higher average salaries.

6.4 Top Five Industries with Maximum Data Science Job Postings

Objective: To identify the industries with the highest number of Data Science-related job postings.

SQL Approach: Grouped records by industry, counted job postings, sorted the results, and selected the top five industries.

Finding: This analysis identifies industries with strong demand for Data Science professionals.

6.5 Companies with Maximum Number of Job Openings

Objective: To identify companies advertising the highest number of job opportunities.

SQL Approach: Grouped job postings by company name and counted the number of openings.

Finding: This analysis highlights companies with comparatively higher recruitment activity.

6.6 Job Titles with Most Number of Jobs

Objective: To identify the most frequently advertised job titles.

SQL Approach: Grouped records by standardized job title and counted the number of postings.

Finding: This analysis identifies the most common job roles in the dataset.

6.7 Salary of Job Titles with Most Number of Jobs

Objective: To compare salaries offered for frequently advertised job titles.

SQL Approach: Grouped job titles and calculated their average salary estimates.

Finding: This analysis helps compare earning potential across popular job roles.

6.8 Skills Required by Companies for Each Job Title

Objective: To identify the technical skills demanded by companies for different job positions.

SQL Approach: Analyzed skill indicator columns such as Python, Spark, AWS, and Excel, where 1 indicates that the skill is required.

Finding: This analysis identifies important technical skills that job seekers should consider developing.

6.9 Relationship Between Average Salary and Education

Objective: To examine the relationship between educational qualifications and salary estimates.

SQL Approach: Grouped records based on the Degree attribute and calculated average salary values.

Finding: This analysis helps understand salary differences across educational qualification categories.

6.10 Additional Feature Analysis

Objective: To analyze other available dataset attributes and identify additional job market patterns.

SQL Approach: Examined relevant attributes such as company ratings, seniority, company size, industry, ownership, and other available features.

Finding: This analysis provides additional information about company characteristics and recruitment patterns.

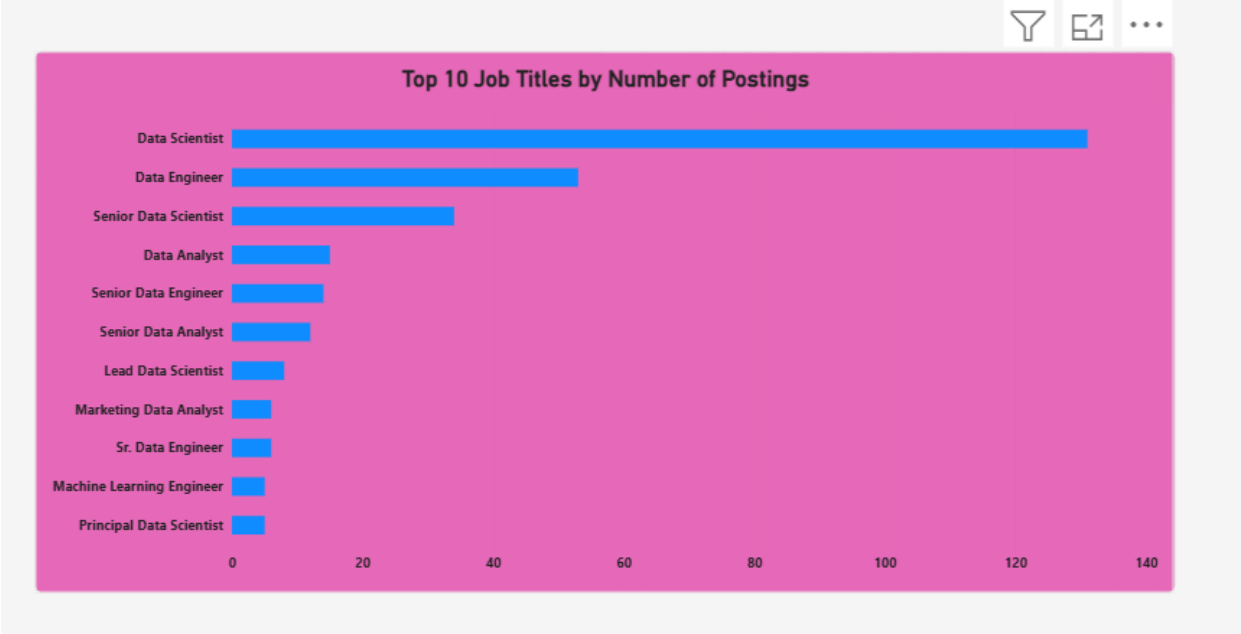
6.11 Overall Job Market Insights

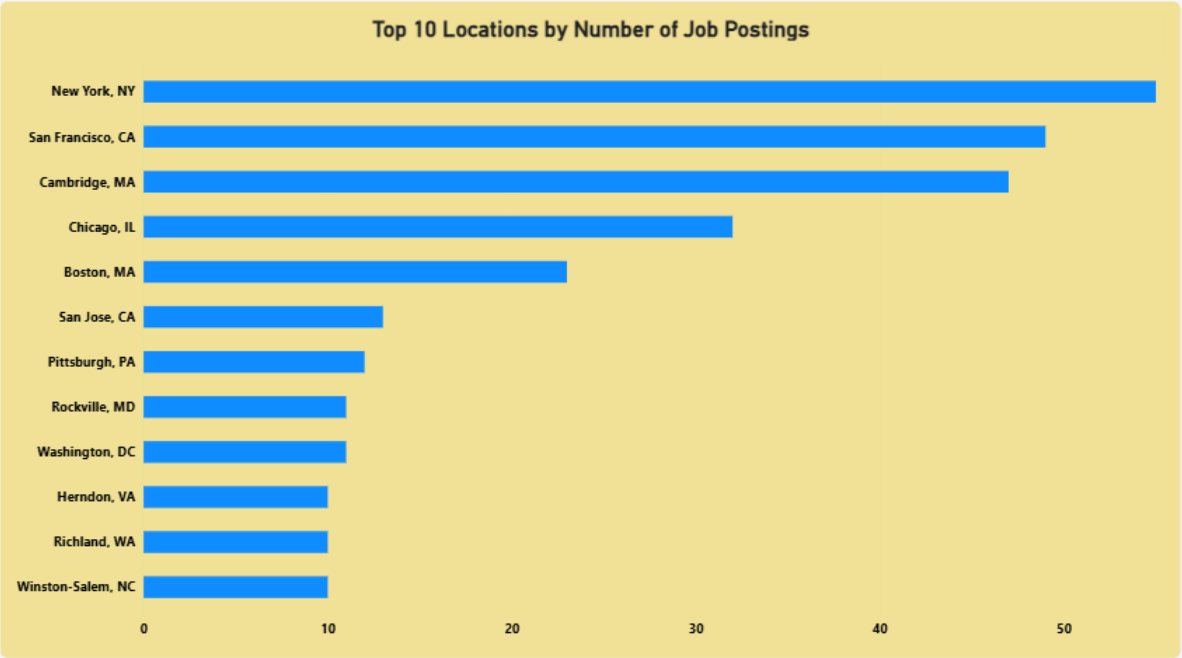
Objective: To combine the results of different analyses and identify meaningful job market trends.

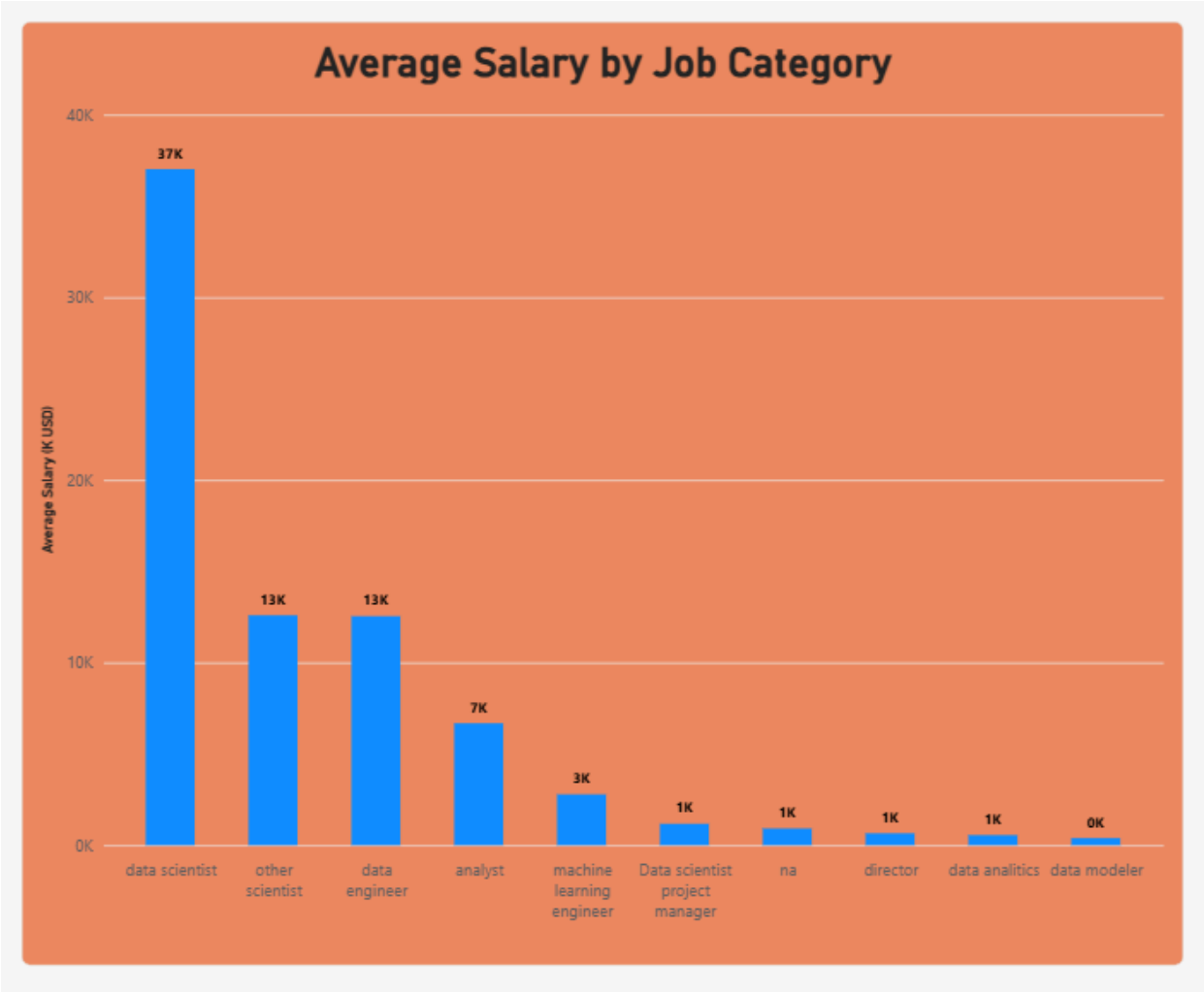
SQL Approach: Compared the outputs of the SQL queries to identify patterns in hiring demand, salaries, locations, companies, industries, and required skills.

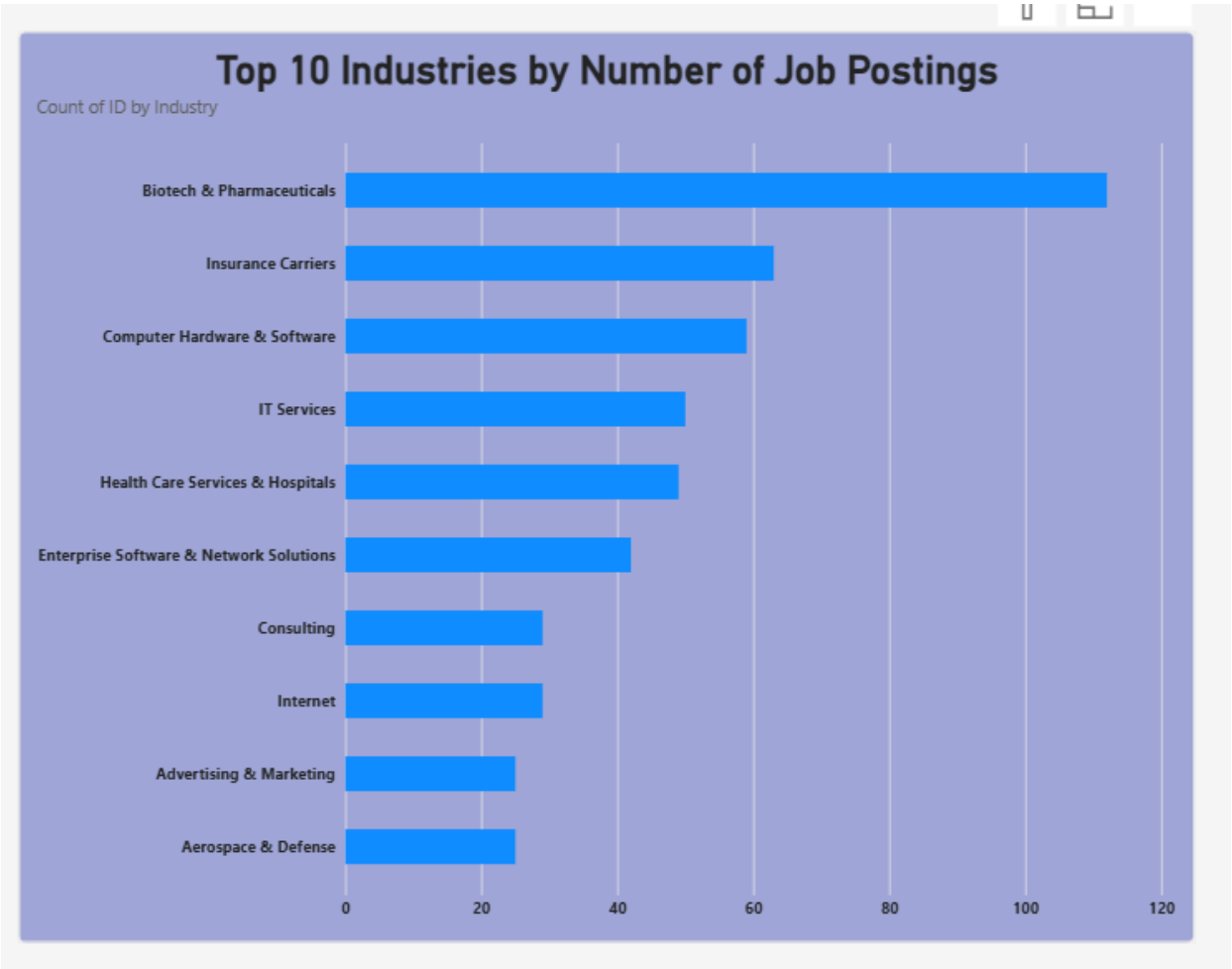
Finding: The combined analysis provides an overall understanding of the Data Science job market.

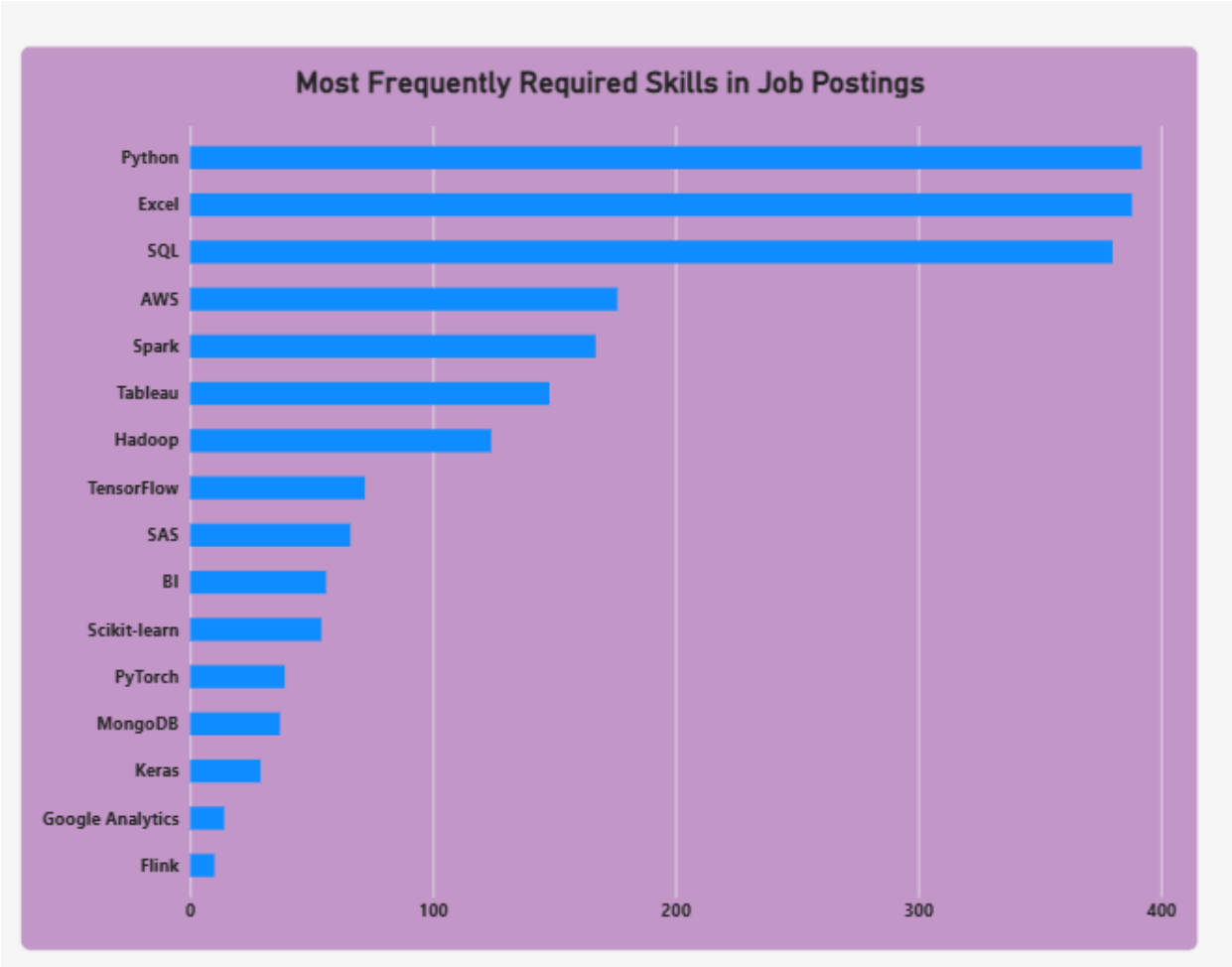
7. DATA VISUALIZATION AND DASHBOARD



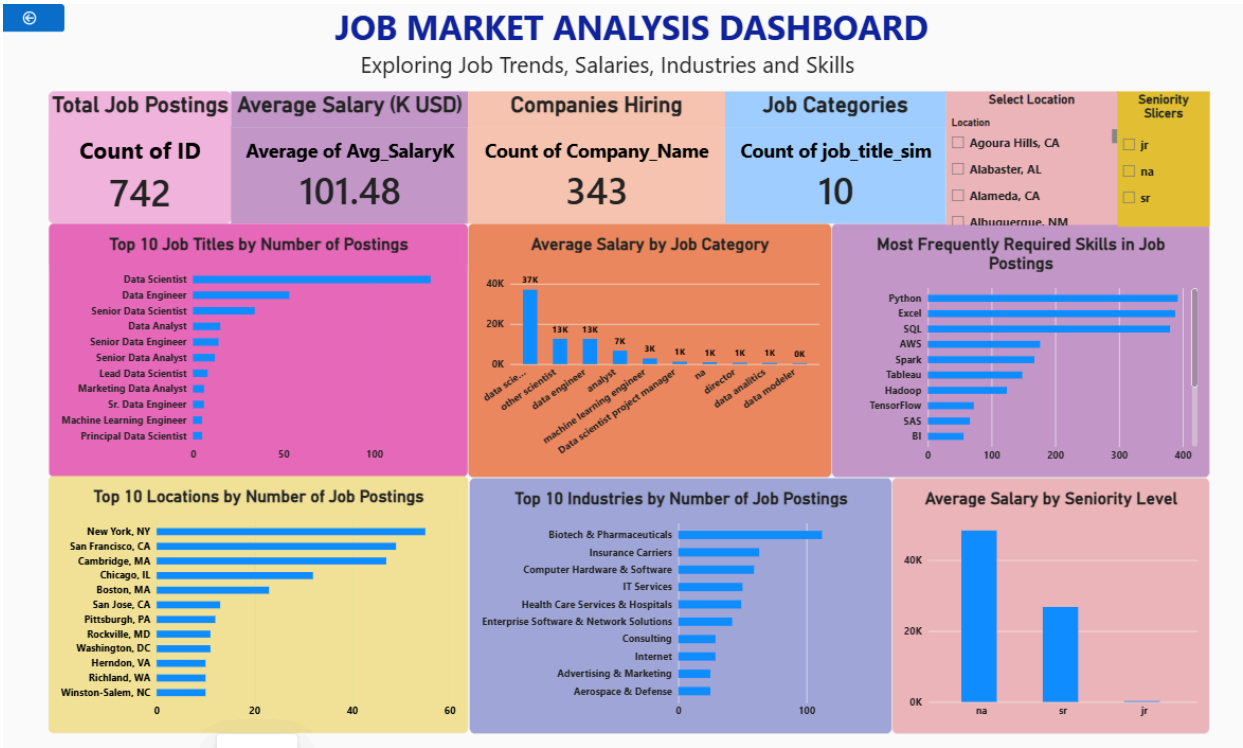

Top 10 Locations by Number of Job Postings
New York, NY
San Francisco, CA
Cambridge, MA
Chicago, IL
Boston, MA
San Jose, CA
Pittsburgh, PA
Rockville, MD
Washington, DC
Herndon, VA
Richland, WA
Winston-Salem, NC
0
10
20
30
40
50


Average Salary by Job Category
40K
30K
20K
10K
0K
Average Salary (K USD)
37K
13K
13K
7K
3K
1K
1K
1K
1K
0K
data scientist
other scientist
data engineer
analyst
machine learning engineer
Data scientist project manager
na
director
data analitics
data modeler


Top 10 Industries by Number of Job Postings
Count of ID by Industry
Biotech & Pharmaceuticals
Insurance Carriers
Computer Hardware & Software
IT Services
Health Care Services & Hospitals
Enterprise Software & Network Solutions
Consulting
Internet
Advertising & Marketing
Aerospace & Defense
0
20
40
60
80
100
120


Most Frequently Required Skills in Job Postings
Python
Excel
SQL
AWS
Spark
Tableau
Hadoop
TensorFlow
SAS
BI
Scikit-learn
PyTorch
MongoDB
Keras
Google Analytics
Flink
0
100
200
300
400


Average Salary by Seniority Level
50K
40K
30K
20K
10K
0K
na
sr
jr



8. KEY INSIGHTS

The SQL analysis and Power BI visualizations provide several useful insights into the Data Science job market.

The major areas of analysis include:

1. **Geographical Demand:** Identifying states with the highest number of job opportunities.
2. **Salary Trends:** Understanding differences in average salary across geographical locations.
3. **Industry Demand:** Identifying industries with the highest number of Data Science-related job postings.
4. **Company Hiring:** Understanding which companies advertise more job openings.
5. **Popular Job Roles:** Identifying the most frequently advertised job titles.

6. **Technical Skills:** Understanding the technical skills most frequently required by employers.
7. **Educational Qualifications:** Examining salary patterns across educational qualification categories.
8. **Company Characteristics:** Understanding how company ratings, size, sectors, and other features contribute to the overall job market analysis.

The findings help job seekers understand employer expectations and assist organizations in understanding recruitment trends.

Note: The specific states, industries, companies, salaries, and skills identified in the charts should be mentioned here using the actual analysis results.

9. RECOMMENDATIONS

Based on the analysis of the job market dataset, the following recommendations are suggested:

Recommendations for Job Seekers:

1. Focus on developing in-demand technical skills such as Python, SQL, Excel, and other skills identified through the analysis.
2. Explore states and locations with higher job posting counts.
3. Compare salary ranges across different job titles before selecting a career path.
4. Improve educational qualifications and practical knowledge according to employer requirements.
5. Build practical projects to demonstrate technical and analytical skills.

Recommendations for Companies:

1. Identify skill gaps and design appropriate employee training programs.
2. Offer competitive salaries based on job roles and geographical market trends.
3. Improve recruitment strategies by focusing on locations with relevant talent availability.
4. Clearly mention technical skills and educational requirements in job descriptions.
5. Use job market analytics to understand recruitment patterns and workforce requirements.

These recommendations can help job seekers make informed career decisions and enable companies to improve their recruitment strategies.

10. CONCLUSION

The Job Market Analysis project provides an overview of employment trends in the Data Science domain using a dataset containing 742 records and 42 features.

SQL was used to retrieve and analyze job-related information, while Microsoft Power BI was used to create interactive visualizations and dashboards.

The project helped examine job opportunities across states, salary distributions, industry demand, company hiring patterns, popular job titles, technical skill requirements, and educational qualifications.

The analysis demonstrates how SQL and Power BI can be used together to convert raw data into meaningful insights.

Overall, this project provides valuable information for job seekers, employers, and organizations to understand job market trends and make better career and recruitment decisions.